How Much Demonstration Data Does Deep Q-Learning from Demonstrations Need?

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Abstract

Deep reinforcement learning (RL) methods often require large amounts of interaction data to achieve strong performance. Deep Q-learning from Demonstrations (DQfD) mitigates this limitation by incorporating demonstrations into the training process, improving early learning efficiency. In this project, we investigate how the quantity of demonstration data affects DQfD performance in two non-Atari environments: Snake and Flappy Bird. We perform a controlled ablation study using up to 50 human demonstration episodes and compare results against a standard Deep Q-Network (DQN) baseline. Our results show that DQfD consistently outperforms DQN in early learning and final performance, while the impact of demonstration quantity varies across environments. Notably, even a single demonstration can improve long-term performance, although it introduces higher variance and instability.

1 Introduction

Deep reinforcement learning has achieved impressive results across a wide range of tasks, but its sample inefficiency remains a major limitation. In many environments, agents require millions of interactions to learn effective policies, which is impractical in real-world scenarios.

Deep Q-learning from Demonstrations (DQfD) addresses this limitation by incorporating expert demonstrations into the learning process, allowing the agent to bootstrap its policy from prior experience. While prior work demonstrated strong results in Atari environments, it did not systematically analyze how the quantity of demonstration data influences performance.

In this project, we investigate the following research question: *How much demonstration data is necessary for DQfD to provide meaningful improvements over standard DQN?* To answer this, we evaluate DQfD on two environments with different characteristics (Snake and Flappy Bird) and analyze performance as a function of demonstration dataset size.

2 Background

Deep Q-Network (DQN): DQN approximates the action-value function using a neural network and learns via temporal-difference updates with experience replay and target networks.

Deep Q-learning from Demonstrations (DQfD): DQfD, proposed by [2], extends DQN by incorporating demonstration data into the replay buffer and introducing additional learning signals. These include:

- A supervised large-margin classification loss to imitate expert actions,
- N-step returns to propagate rewards more efficiently,
- Prioritized experience replay with higher sampling priority for demonstration data.

This combination enables faster early learning and improved sample efficiency.

3 Methodology

3.1 Environments

We evaluate our methods on two Gym environments:

Snake:

- Observation space: 6-dimensional feature vector (food direction and collision risk)
- Action space: 4 discrete actions (up, down, left, right)

Flappy Bird:

- Observation space: 180-dimensional ray-based vector (downsampled to 90)
- Action space: 2 discrete actions (flap or no-op)

These environments differ in complexity: Snake requires planning, while Flappy Bird requires precise timing.

3.2 Demonstration Data

Human gameplay was recorded to generate demonstration datasets. For each environment, we collected up to 50 episodes. The gameplay was produced by non-experts, so it includes suboptimal behaviors and variability in quality.

To evaluate the effect of demonstration quantity, we trained models using: 0, 1, 10, 25, 50 demonstration episodes. Demonstrations were randomly sampled for each training run, ensuring variability across seeds.

3.3 Model and Training Setup

We implemented both DQN and DQfD using multilayer perceptrons.

Key implementation details:

- Replay buffer with prioritized sampling and permanent storage of demonstrations
- N-step returns with $n = 10$
- Large-margin supervised loss for demonstration samples
- Pretraining phase using only demonstration data
- Double DQN updates

The total DQfD loss is:

$$\mathcal{L} = \lambda_{DQ}\mathcal{L}_{1-step} + \lambda_n\mathcal{L}_{n-step} + \lambda_{JE}\mathcal{L}_{margin} + \lambda_{L2}\|\theta\|^2$$

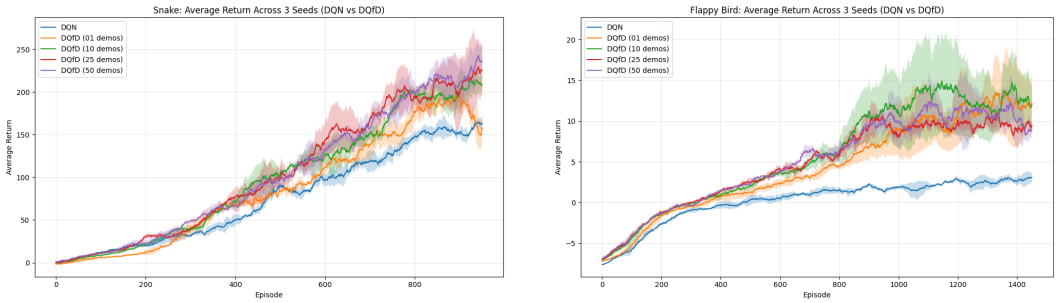
Each configuration was trained using 3 different random seeds for robustness.

3.4 Evaluation Metrics

- **Learning curves:** Average episodic return as a function of training episodes, which captures learning dynamics and sample efficiency over time.
- **Initial performance:** Mean return over the first 100 training episodes, used to assess early-stage learning and the immediate impact of demonstration data.
- **Final performance:** Mean return over the last 100 training episodes, which summarizes performance after training has progressed.

4 Experimental Results

4.1 Learning Curves



(a) Snake (b) Flappy Bird

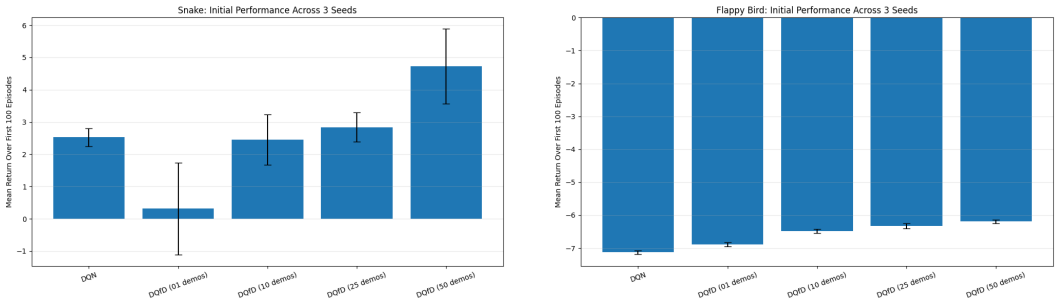
Figure 1: Average return vs. training episodes for DQN and DQfD with varying numbers of demonstrations.

Figure 1 shows the learning dynamics across both environments. All models exhibit learning behavior, but DQfD consistently outperforms DQN.

In Snake (Figure 1a), the 1-demonstration model performs worst during early training, likely due to overfitting to a single trajectory. However, it eventually surpasses DQN, indicating that even limited demonstrations provide useful structural guidance.

In Flappy Bird (Figure 1b), the model with the fewest demonstrations initially performs worst but gradually improves and eventually surpasses other models during later training, suggesting strong long-term learning capability even with minimal prior data.

4.2 Initial Performance



(a) Snake (b) Flappy Bird

Figure 2: Average return during the first 100 episodes.

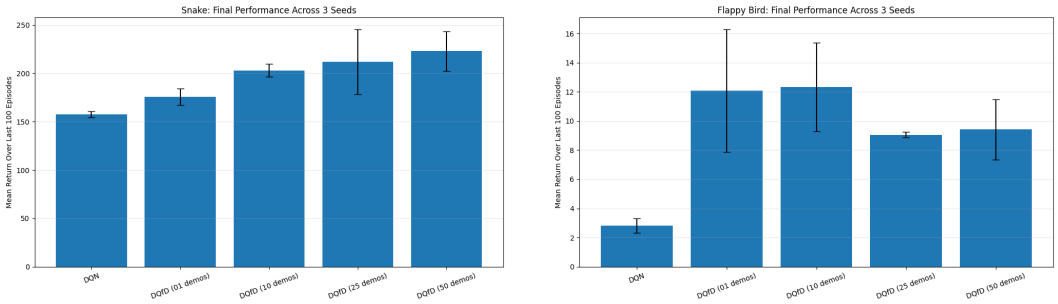
Figure 2 highlights early-stage performance.

In Snake (Figure 2a), performance improves with the number of demonstrations, with 50 demonstrations achieving the best results. The 1-demonstration setting shows high variance and underperforms DQN, indicating instability when relying on minimal data.

Similarly, in Flappy Bird (Figure 2b), performance increases with the number of demonstrations, and DQN exhibits the worst performance across all models.

4.3 Final Performance

Figure 3 shows final performance after training convergence.



(a) Snake (b) Flappy Bird

Figure 3: Average return during the last 100 episodes.

In Snake (Figure 3a), DQfD outperforms DQN across all configurations, and there is a clear positive correlation between the number of demonstrations and final performance.

In contrast, Flappy Bird (Figure 3b) does not exhibit a clear trend with respect to demonstration quantity. The 1-demonstration model achieves the highest mean reward but also shows the highest variance, while larger datasets yield more stable performance.

5 Conclusion and Future Work

This project demonstrates that DQfD consistently improves both sample efficiency and final performance compared to a standard DQN baseline across different environments.

Key findings:

- Demonstrations improve learning compared to the baseline in early and final stages.
- Increasing the number of demonstrations generally leads to improved performance. This trend is most evident in early-stage learning for Flappy Bird, and in later-stage performance for Snake.
- Even a single demonstration can be beneficial. However, it introduces instability due to sensitivity to the quality of the provided example.
- The impact of demonstration quantity is environment-dependent: while similar trends are observed, the magnitude and consistency of improvements vary between Snake and Flappy Bird.
- DQfD configurations tend to exhibit higher variance than DQN, likely due to the stochastic selection of demonstrations and the presence of suboptimal (non-expert) trajectories in the dataset.

Future Work:

An important extension of this work is to analyze the role of **demonstration quality** in addition to quantity. Specifically, comparing curated high-quality demonstrations against low-quality or noisy ones would help determine whether performance gains are primarily driven by the amount of data or its quality.

Such an analysis would provide deeper insight into the robustness of DQfD and inform strategies for efficiently collecting or filtering demonstration data in practical applications.

6 Contributions and Acknowledgements

Miguel Carrillo conducted all aspects of the project. Code and results are available at `https://github.com/ma-carrillo/dqfd-demonstration-ablation` [1].

This project builds upon the Deep Q-learning from Demonstrations (DQfD) framework proposed by [2]. The experiments were conducted using open-source Gym-based environments for Snake [4] and Flappy Bird [3], whose developers are gratefully acknowledged.

References

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